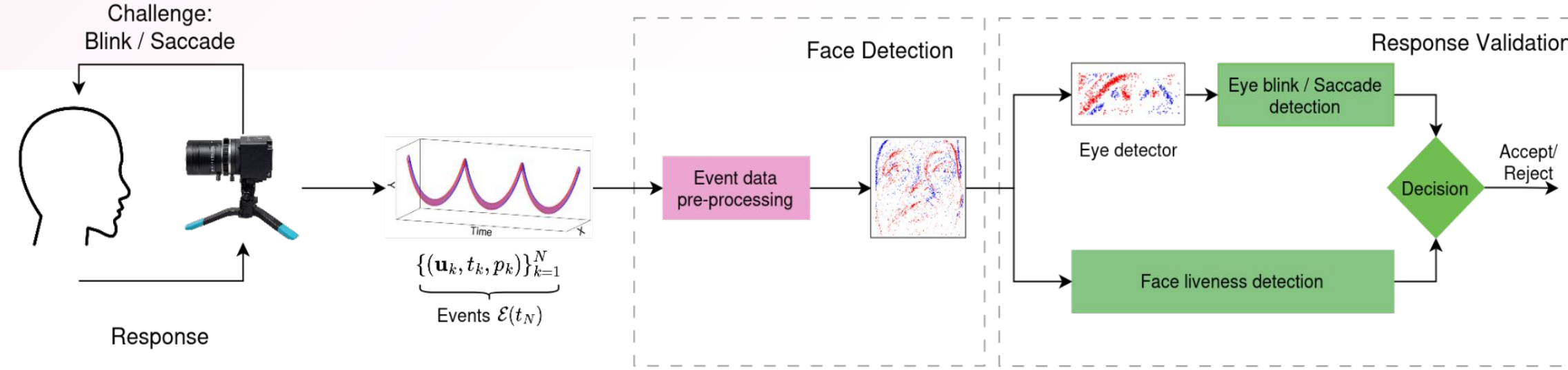


Event-based Liveness Detection using Temporal Ocular Dynamics: An Exploratory Approach

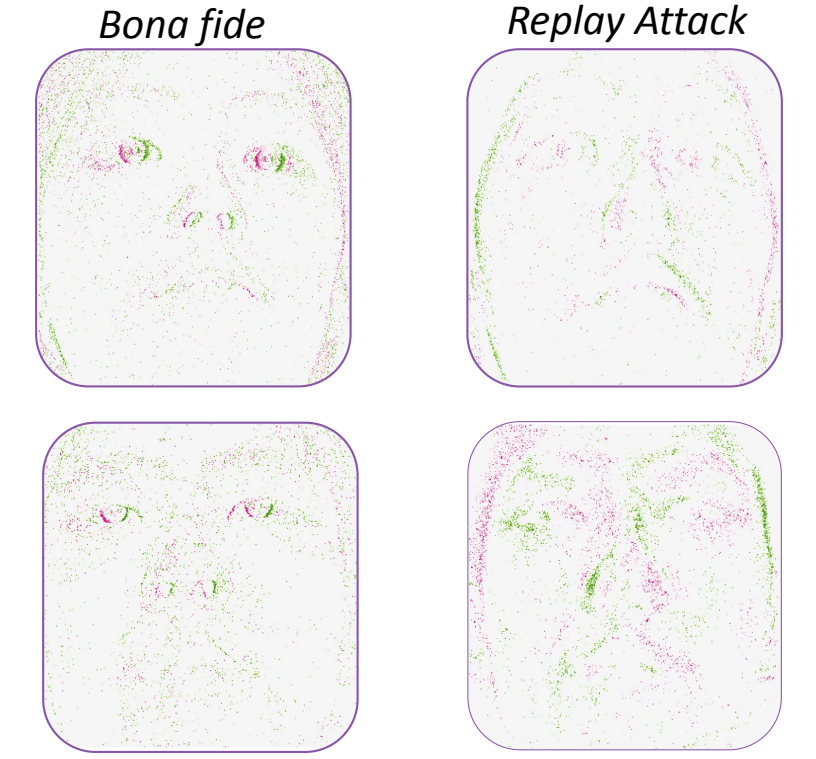
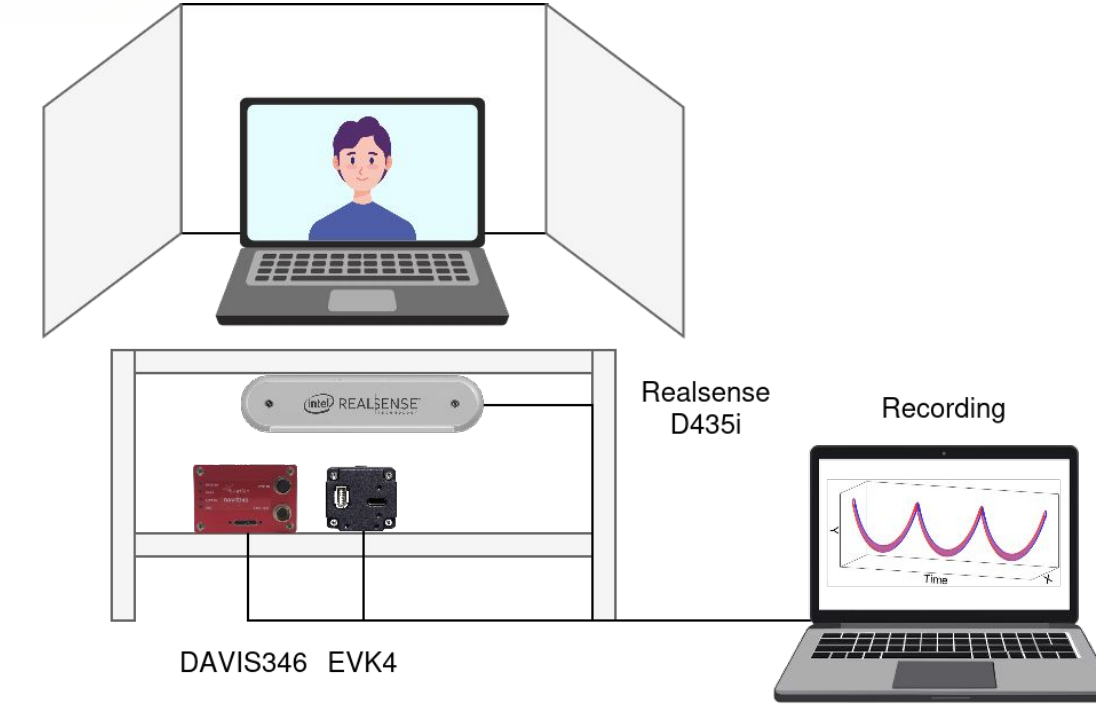
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We present a replay attack dataset, derived from RGBE-Gaze, designed to study event-based liveness detection

We propose an event-based method for the temporal segmentation of saccades and blinks in remote-eye settings



Event-based liveness detection pipeline using eye movements as a challenge-response mechanism to enable real-time response verification



Comparison of bona fide and replay attack clips using Surface of Active Events

I. Face Liveness

Given a facial video, determine whether the face is genuine or a spoof

RGB Challenges: Cross-dataset generalization, diverse attacks and sensors, and in-the-wild scenarios

Why event cameras?

In a *replay attack*, the sampling process and other screen artifacts captured by the camera generate an event signal that is distinguishable from a genuine one. Subtle eye movements could be used as liveness cues

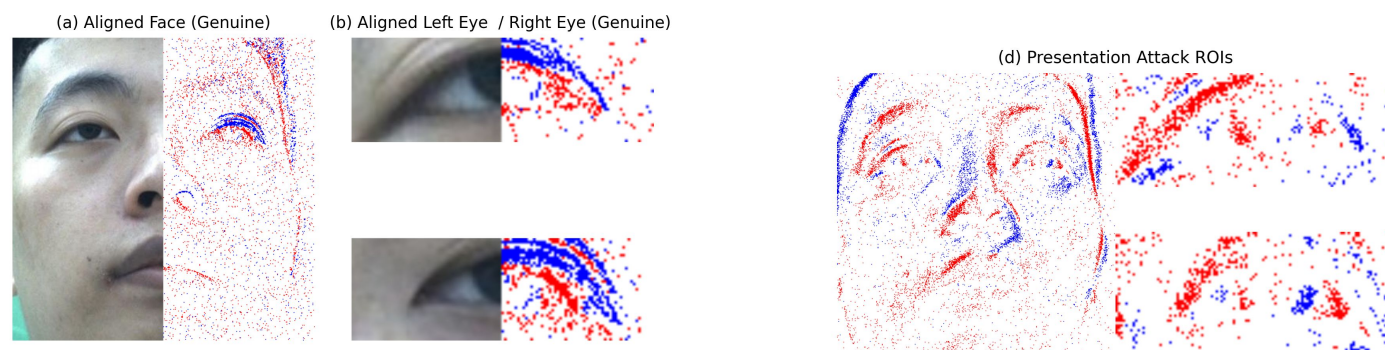
II. Remote-Eye Scenario

Subjects are at a distance ≥ 50 cm

Challenge-response: Segment the onset and offset of fast ocular movements (saccades and blinks)

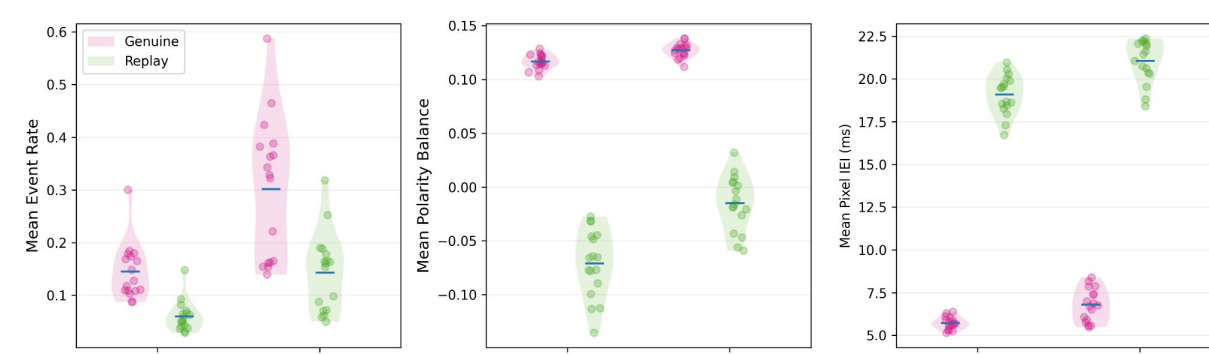
III. Dataset RGBE + Replay Attack

Our dataset extends the RGBE-Gaze[1] with events and RGB replay attack samples



	Genuine		Replay	
	Events	RGB	Events	RGB
Device	EVK4	FLIR	EVK4	RealSense
FPS	N/A	50	N/A	30
#Subjects	17	17	17	17

Display-induced artifacts

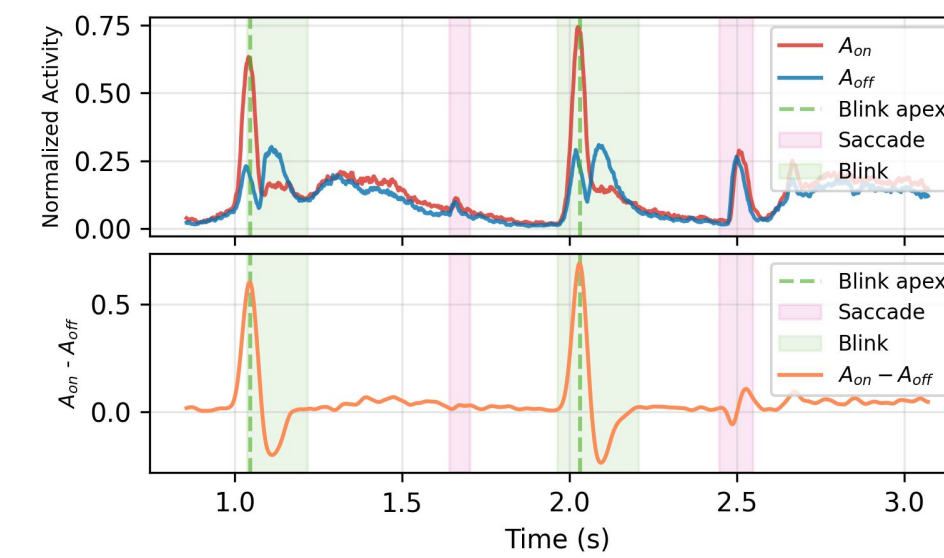


These statistical features suggest a clear class separation; however, we argue that they may be insufficient for building a robust detection system

V. Saccades and Blink Segmentation

- It is possible to characterize a temporal signature of blinks under the activity profile

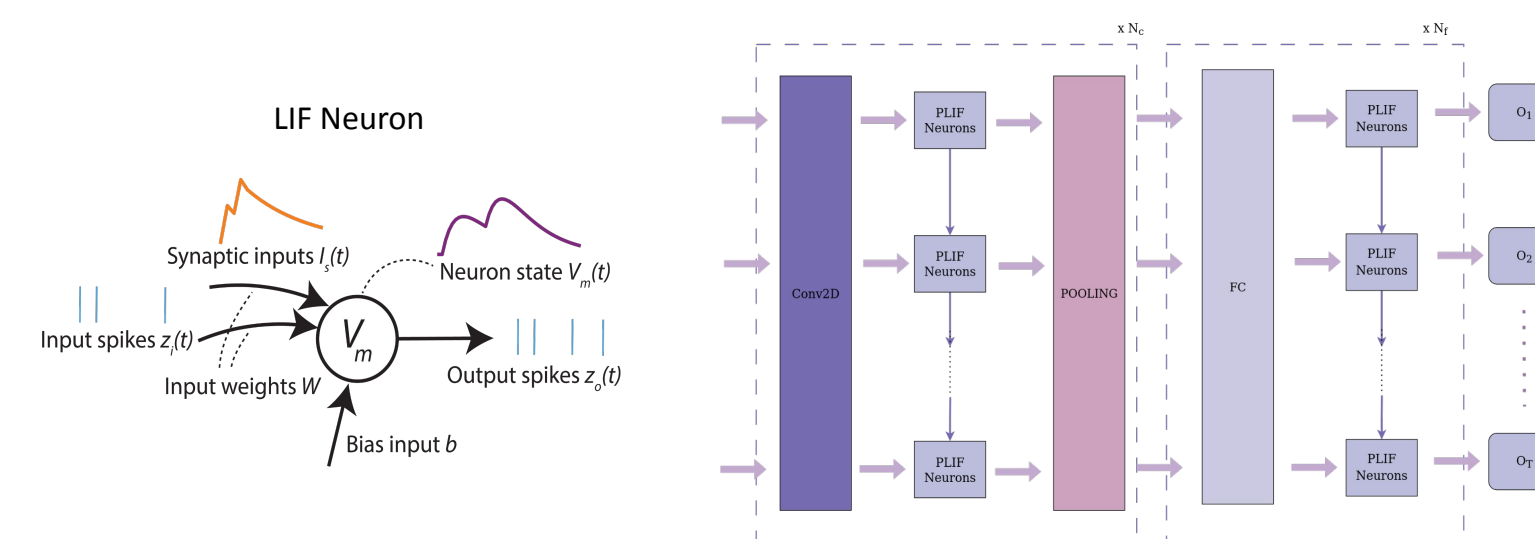
$$A_p(t_i) = A_p(t_{u_p}) e^{-\frac{t_i - t_{u_p}}{\tau}} + \frac{1}{\mu} \mathbb{I}[p_i = p]$$



- The proposed temporal blink segmentation method consists of finding a positive prominent peak followed by a negative peak in the difference of the on and off signal
- Saccades do not exhibit the same temporal signature and can therefore be confused with other transient motion artifacts, such as slight head movements. We propose a Temporal Convolutional Network (TCN) as baseline

VI. Liveness Detection

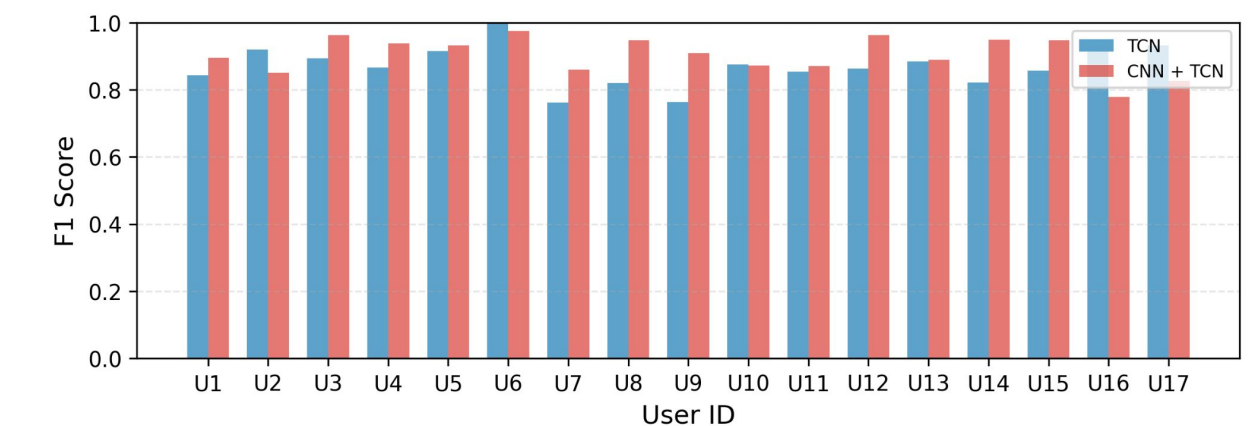
- Binary classification of clips: Genuine vs Replay Attack
- Models: Vanilla ViT-B16 using SAE frames vs Spiking Convolutional Neural Network



Results

Segmentation	Method	Precision	Recall	F1
	Peaks	64.63	60.92	62.72
Saccade	TCN	89.65	89.65	89.65
Blink	Peaks	97.62	93.18	95.35

We evaluate the results using an IoU threshold of 0.5. The method based on Activity Profile signal analysis achieves good results only for blinks



Using LOSO-CV, we observe that incorporating the spatial information of events through the Surface of Active Events (SAE) improves the average F1-score by 5%

Results

	SCNN	SCNN	ViT	ViT
	Full Events	100 ms Events	RGB Frames	100 ms SAE
Backbone	Conv.	Conv.	B16	B16
Top-1 Accuracy (%)	95.37	92.48	92.15	90.62
APCER (%)	4.20	6.90	8.20	10.20
BPCER (%)	5.10	8.10	7.60	8.50
ACER (%)	4.65	7.50	7.90	9.35

Using a voxel-grid representation of events, the SCNN without temporal window partitioning achieved the best results

Conclusions

- Although the SCNN based method achieved promising results for liveness detection, its generalization to other attack types and instruments needs to be further evaluated
- Temporal modeling of event signals enabled accurate segmentation of saccades and blinks, supporting its use as mechanism for dynamic liveness detection

